

# Звіт з лабораторної роботи №5.1

Роботу виконав студент групи МІТ-31

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## Хід роботи

### Завдання 1. Створення колекцій

```
PS C:\Users\Lenovo> mongosh
Current Mongosh Log ID: 6a052bd1b52d73bb80abc113
Connecting to:      mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.8.3
Using MongoDB:      8.3.2
Using Mongosh:       2.8.3

For mongosh info see: https://www.mongodb.com/docs/mongosh-shell/

To help improve our products, anonymous usage data is collected and sent to MongoDB periodically (https://www.mongodb.com/legal/privacy-policy).
You can opt-out by running the disableTelemetry() command.

-----
The server generated these startup warnings when booting
2026-05-14T04:52:09.564+03:00: Access control is not enabled for the database. Read and write access to data and configuration is unrestricted
-----

test> use lab5
switched to db lab5
lab5> db.createCollection("members")
| db.createCollection("classes")
| db.createCollection("visits")
{ ok: 1 }
lab5> show collections
classes
members
visits
```

### Завдання 2 - CRUD: Create. Наповнення бази даними

```
visits
lab5> db.members.insertMany([
|   { name: "Олександр Коваленко", email: "alex@test.com", plan: "Basic" },
|   { name: "Марія Ткаченко", email: "maria@test.com", plan: "Premium" },
|   { name: "Іван Петренко", email: "ivan@test.com", plan: "VIP" },
|   { name: "Анна Бойко", email: "anna@test.com", plan: "Basic" },
|   { name: "Сергій Мельник", email: "serg@test.com", plan: "Premium" },
|   { name: "Олена Шевченко", email: "olena@test.com", plan: "VIP" },
|   { name: "Дмитро Лисенко", email: "dmytro@test.com", plan: "Basic" },
|   { name: "Юлія Кравченко", email: "yulia@test.com", plan: "Premium" },
|   { name: "Максим Романенко", email: "max@test.com", plan: "Basic" },
|   { name: "Вікторія Павленко", email: "vika@test.com", plan: "VIP" }
| ])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6a052c4eb52d73bb80abc114'),
    '1': ObjectId('6a052c4eb52d73bb80abc115'),
    '2': ObjectId('6a052c4eb52d73bb80abc116'),
    '3': ObjectId('6a052c4eb52d73bb80abc117'),
    '4': ObjectId('6a052c4eb52d73bb80abc118'),
    '5': ObjectId('6a052c4eb52d73bb80abc119'),
    '6': ObjectId('6a052c4eb52d73bb80abc11a'),
    '7': ObjectId('6a052c4eb52d73bb80abc11b'),
    '8': ObjectId('6a052c4eb52d73bb80abc11c'),
    '9': ObjectId('6a052c4eb52d73bb80abc11d')
  }
}
lab5> |
```

```

lab5> db.classes.insertMany([
|   { title: "Yoga", level: "Beginner", price: 150 },
|   { title: "Yoga", level: "Advanced", price: 250 },
|   { title: "Pilates", level: "Intermediate", price: 200 },
|   { title: "CrossFit", level: "Advanced", price: 300 },
|   { title: "Zumba", level: "Beginner", price: 120 },
|   { title: "Boxing", level: "Intermediate", price: 220 },
|   { title: "Stretching", level: "Beginner", price: 100 },
|   { title: "TRX", level: "Intermediate", price: 250 },
|   { title: "Cycling", level: "Beginner", price: 180 },
|   { title: "Aqua Aerobics", level: "Intermediate", price: 200 },
|   { title: "Step Aerobics", level: "Beginner", price: 150 },
|   { title: "BodyPump", level: "Advanced", price: 280 },
|   { title: "MMA", level: "Advanced", price: 350 },
|   { title: "Karate", level: "Beginner", price: 160 },
|   { title: "Dance", level: "Beginner", price: 130 }
| ])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6a052c71b52d73bb80abc11e'),
    '1': ObjectId('6a052c71b52d73bb80abc11f'),
    '2': ObjectId('6a052c71b52d73bb80abc120'),
    '3': ObjectId('6a052c71b52d73bb80abc121'),
    '4': ObjectId('6a052c71b52d73bb80abc122'),
    '5': ObjectId('6a052c71b52d73bb80abc123'),
    '6': ObjectId('6a052c71b52d73bb80abc124'),
    '7': ObjectId('6a052c71b52d73bb80abc125'),
    '8': ObjectId('6a052c71b52d73bb80abc126'),
    '9': ObjectId('6a052c71b52d73bb80abc127'),
    '10': ObjectId('6a052c71b52d73bb80abc128'),
    '11': ObjectId('6a052c71b52d73bb80abc129'),
    '12': ObjectId('6a052c71b52d73bb80abc12a'),
    '13': ObjectId('6a052c71b52d73bb80abc12b'),
    '14': ObjectId('6a052c71b52d73bb80abc12c')
  }
}
lab5> |

```

```

lab5> let visitsArray = [];
let statuses = ["Completed", "Cancelled", "No-show"];
let plans = ["Basic", "Premium", "VIP"];
for(let i = 1; i <= 30; i++) {
  visitsArray.push({
    visitNo: "V" + String(i).padStart(3, '0'),
    memberEmail: "alex@test.com",
    date: new Date(`2026-05-${Math.floor(Math.random() * 28) + 1}`),
    items: [
      { class: "Yoga", qty: 1, price: 150 },
      { class: "Stretching", qty: 1, price: 100 }
    ],
    status: statuses[Math.floor(Math.random() * statuses.length)]
  });
}
db.visits.insertMany(visitsArray);

```

```

{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6a052c9fb52d73bb80abc12d'),
    '1': ObjectId('6a052c9fb52d73bb80abc12e'),
    '2': ObjectId('6a052c9fb52d73bb80abc12f'),
    '3': ObjectId('6a052c9fb52d73bb80abc130'),
    '4': ObjectId('6a052c9fb52d73bb80abc131'),
    '5': ObjectId('6a052c9fb52d73bb80abc132'),
    '6': ObjectId('6a052c9fb52d73bb80abc133'),
    '7': ObjectId('6a052c9fb52d73bb80abc134'),
    '8': ObjectId('6a052c9fb52d73bb80abc135'),
    '9': ObjectId('6a052c9fb52d73bb80abc136'),
    '10': ObjectId('6a052c9fb52d73bb80abc137'),
    '11': ObjectId('6a052c9fb52d73bb80abc138'),
    '12': ObjectId('6a052c9fb52d73bb80abc139'),
    '13': ObjectId('6a052c9fb52d73bb80abc13a'),
    '14': ObjectId('6a052c9fb52d73bb80abc13b'),
    '15': ObjectId('6a052c9fb52d73bb80abc13c'),
    '16': ObjectId('6a052c9fb52d73bb80abc13d'),
    '17': ObjectId('6a052c9fb52d73bb80abc13e'),
    '18': ObjectId('6a052c9fb52d73bb80abc13f'),
    '19': ObjectId('6a052c9fb52d73bb80abc140'),
    '20': ObjectId('6a052c9fb52d73bb80abc141'),
    '21': ObjectId('6a052c9fb52d73bb80abc142'),
    '22': ObjectId('6a052c9fb52d73bb80abc143'),
    '23': ObjectId('6a052c9fb52d73bb80abc144'),
    '24': ObjectId('6a052c9fb52d73bb80abc145'),
    '25': ObjectId('6a052c9fb52d73bb80abc146'),
    '26': ObjectId('6a052c9fb52d73bb80abc147'),
    '27': ObjectId('6a052c9fb52d73bb80abc148'),
    '28': ObjectId('6a052c9fb52d73bb80abc149'),
    '29': ObjectId('6a052c9fb52d73bb80abc14a')
  }
}

```

## Базові операції: Читання, Оновлення, Видалення (Завдання 2)

```

lab5> db.members.find({ plan: "Premium" })
[
  {
    _id: ObjectId('6a052c4eb52d73bb80abc115'),
    name: 'Марія Ткаченко',
    email: 'maria@test.com',
    plan: 'Premium'
  },
  {
    _id: ObjectId('6a052c4eb52d73bb80abc118'),
    name: 'Сергій Мельник',
    email: 'serg@test.com',
    plan: 'Premium'
  },
  {
    _id: ObjectId('6a052c4eb52d73bb80abc11b'),
    name: 'Юлія Кравченко',
    email: 'yulia@test.com',
    plan: 'Premium'
  }
]

```

```

lab5> // Змінюємо план одного клієнта (оновлюємо лише одне поле через $set)
| db.members.updateOne({ email: "alex@test.com" }, { $set: { plan: "VIP" } })
|
| // Збільшуємо ціну на 20 грн для всіх початкових класів (через $inc)
| db.classes.updateMany({ level: "Beginner" }, { $inc: { price: 20 } })
| {
|   acknowledged: true,
|   insertedId: null,
|   matchedCount: 7,
|   modifiedCount: 7,
|   upsertedCount: 0
| }
lab5> |

```

```

lab5> db.visits.deleteMany({ status: "Cancelled" })
{ acknowledged: true, deletedCount: 15 }
lab5> |

```

### Завдання 3. Запити та фільтрація

```

lab5> db.visits.deleteMany({ status: "Cancelled" })
{ acknowledged: true, deletedCount: 15 }
lab5> db.classes.find({ price: { $gt: 200 } })
[
  {
    _id: ObjectId('6a052c71b52d73bb80abc11f'),
    title: 'Yoga',
    level: 'Advanced',
    price: 250
  },
  {
    _id: ObjectId('6a052c71b52d73bb80abc121'),
    title: 'CrossFit',
    level: 'Advanced',
    price: 300
  },
  {
    _id: ObjectId('6a052c71b52d73bb80abc123'),
    title: 'Boxing',
    level: 'Intermediate',
    price: 220
  },
  {
    _id: ObjectId('6a052c71b52d73bb80abc125'),
    title: 'TRX',
    level: 'Intermediate',
    price: 250
  },
  {
    _id: ObjectId('6a052c71b52d73bb80abc129'),
    title: 'BodyPump',
    level: 'Advanced',
    price: 280
  },
  {
    _id: ObjectId('6a052c71b52d73bb80abc12a'),
    title: 'MMA',
    level: 'Advanced',
    price: 350
  }
]

```

```
lab5> db.members.find({ plan: { $in: ["Basic", "Premium"] } })
[
  {
    _id: ObjectId('6a052c4eb52d73bb80abc115'),
    name: 'Марія Ткаченко',
    email: 'maria@test.com',
    plan: 'Premium'
  },
  {
    _id: ObjectId('6a052c4eb52d73bb80abc117'),
    name: 'Анна Бойко',
    email: 'anna@test.com',
    plan: 'Basic'
  },
  {
    _id: ObjectId('6a052c4eb52d73bb80abc118'),
    name: 'Сергій Мельник',
    email: 'serg@test.com',
    plan: 'Premium'
  },
  {
    _id: ObjectId('6a052c4eb52d73bb80abc11a'),
    name: 'Дмитро Лисенко',
    email: 'dmytro@test.com',
    plan: 'Basic'
  },
  {
    _id: ObjectId('6a052c4eb52d73bb80abc11b'),
    name: 'Юлія Кравченко',
    email: 'yulia@test.com',
    plan: 'Premium'
  },
  {
    _id: ObjectId('6a052c4eb52d73bb80abc11c'),
    name: 'Максим Романенко',
    email: 'max@test.com',
    plan: 'Basic'
  }
]
```

```

lab5> db.classes.find({
|   $and: [
|     { level: "Advanced" },
|     { price: { $lte: 300 } }
|   ]
| })
[
|   {
|     _id: ObjectId('6a052c71b52d73bb80abc11f'),
|     title: 'Yoga',
|     level: 'Advanced',
|     price: 250
|   },
|   {
|     _id: ObjectId('6a052c71b52d73bb80abc121'),
|     title: 'CrossFit',
|     level: 'Advanced',
|     price: 300
|   },
|   {
|     _id: ObjectId('6a052c71b52d73bb80abc129'),
|     title: 'BodyPump',
|     level: 'Advanced',
|     price: 280
|   }
| ]
lab5> |

```

```

lab5> db.visits.find({
|   $or: [
|     { status: "No-show" },
|     { date: { $lt: new Date("2026-05-15") } }
|   ]
| })
[
|   {
|     _id: ObjectId('6a052c9fb52d73bb80abc12e'),
|     visitNo: 'V002',
|     memberEmail: 'alex@test.com',
|     date: ISODate('2026-05-02T21:00:00.000Z'),
|     items: [
|       { class: 'Yoga', qty: 1, price: 150 },
|       { class: 'Stretching', qty: 1, price: 100 }
|     ],
|     status: 'Completed'
|   },
|   {
|     _id: ObjectId('6a052c9fb52d73bb80abc12f'),
|     visitNo: 'V003',
|     memberEmail: 'alex@test.com',
|     date: ISODate('2026-05-07T21:00:00.000Z'),
|     items: [
|       { class: 'Yoga', qty: 1, price: 150 },
|       { class: 'Stretching', qty: 1, price: 100 }
|     ],
|     status: 'Completed'
|   },
|   {
|     _id: ObjectId('6a052c9fb52d73bb80abc130'),
|     visitNo: 'V004',
|     memberEmail: 'alex@test.com',
|     date: ISODate('2026-05-10T00:00:00.000Z'),
|     items: [
|       { class: 'Yoga', qty: 1, price: 150 },
|       { class: 'Stretching', qty: 1, price: 100 }
|     ],
|     status: 'No-show'
|   }
| ]

```

```

    status: 'No-show'
  },
  {
    _id: ObjectId('6a052c9fb52d73bb80abc131'),
    visitNo: 'V005',
    memberEmail: 'alex@test.com',
    date: ISODate('2026-05-11T00:00:00.000Z'),
    items: [
      { class: 'Yoga', qty: 1, price: 150 },
      { class: 'Stretching', qty: 1, price: 100 }
    ],
    status: 'Completed'
  },
  {
    _id: ObjectId('6a052c9fb52d73bb80abc132'),
    visitNo: 'V006',
    memberEmail: 'alex@test.com',
    date: ISODate('2026-04-30T21:00:00.000Z'),
    items: [
      { class: 'Yoga', qty: 1, price: 150 },
      { class: 'Stretching', qty: 1, price: 100 }
    ],
    status: 'Completed'
  },
  {
    _id: ObjectId('6a052c9fb52d73bb80abc136'),
    visitNo: 'V010',
    memberEmail: 'alex@test.com',
    date: ISODate('2026-05-16T00:00:00.000Z'),
    items: [
      { class: 'Yoga', qty: 1, price: 150 },
      { class: 'Stretching', qty: 1, price: 100 }
    ],
    status: 'No-show'
  },
  {
    _id: ObjectId('6a052c9fb52d73bb80abc138'),
    visitNo: 'V012',
    memberEmail: 'alex@test.com',
    date: ISODate('2026-05-07T21:00:00.000Z'),

```

```

lab5> db.visits.find({ "items.price": { $gte: 150 } })
[
  {
    _id: ObjectId('6a052c9fb52d73bb80abc12e'),
    visitNo: 'V002',
    memberEmail: 'alex@test.com',
    date: ISODate('2026-05-02T21:00:00.000Z'),
    items: [
      { class: 'Yoga', qty: 1, price: 150 },
      { class: 'Stretching', qty: 1, price: 100 }
    ],
    status: 'Completed'
  },
  {
    _id: ObjectId('6a052c9fb52d73bb80abc12f'),
    visitNo: 'V003',
    memberEmail: 'alex@test.com',
    date: ISODate('2026-05-07T21:00:00.000Z'),
    items: [
      { class: 'Yoga', qty: 1, price: 150 },
      { class: 'Stretching', qty: 1, price: 100 }
    ],
    status: 'Completed'
  },
  {
    _id: ObjectId('6a052c9fb52d73bb80abc130'),
    visitNo: 'V004',
    memberEmail: 'alex@test.com',
    date: ISODate('2026-05-10T00:00:00.000Z'),
    items: [
      { class: 'Yoga', qty: 1, price: 150 },
      { class: 'Stretching', qty: 1, price: 100 }
    ],
    status: 'No-show'
  },
  {
    _id: ObjectId('6a052c9fb52d73bb80abc131'),
    visitNo: 'V005',
    memberEmail: 'alex@test.com',
    date: ISODate('2026-05-11T00:00:00.000Z'),

```

```

    _id: ObjectId('6a052c9fb52d73bb80abc131'),
    visitNo: 'V005',
    memberEmail: 'alex@test.com',
    date: ISODate('2026-05-11T00:00:00.000Z'),
    items: [
      { class: 'Yoga', qty: 1, price: 150 },
      { class: 'Stretching', qty: 1, price: 100 }
    ],
    status: 'Completed'
  },
  {
    _id: ObjectId('6a052c9fb52d73bb80abc132'),
    visitNo: 'V006',
    memberEmail: 'alex@test.com',
    date: ISODate('2026-04-30T21:00:00.000Z'),
    items: [
      { class: 'Yoga', qty: 1, price: 150 },
      { class: 'Stretching', qty: 1, price: 100 }
    ],
    status: 'Completed'
  },
  {
    _id: ObjectId('6a052c9fb52d73bb80abc133'),
    visitNo: 'V007',
    memberEmail: 'alex@test.com',
    date: ISODate('2026-05-21T00:00:00.000Z'),
    items: [
      { class: 'Yoga', qty: 1, price: 150 },
      { class: 'Stretching', qty: 1, price: 100 }
    ],
    status: 'Completed'
  },
  {
    _id: ObjectId('6a052c9fb52d73bb80abc136'),
    visitNo: 'V010',
    memberEmail: 'alex@test.com',
    date: ISODate('2026-05-16T00:00:00.000Z'),
    items: [
      { class: 'Yoga', qty: 1, price: 150 },
      { class: 'Stretching', qty: 1, price: 100 }
    ]
  }
]

```

#### Завдання 4. Індексація

```

lab5> // Унікальний індекс для email клієнтів (щоб не було дублікатів)
| db.members.createIndex({ email: 1 }, { unique: true })
|
| // Складений індекс для занять (рівень + ціна)
| db.classes.createIndex({ level: 1, price: 1 })
|
| // Складений індекс для відвідувань (статус + дата за спаданням)
| db.visits.createIndex({ status: 1, date: -1 })
status_1_date_-1

```



```

lab5> db.visits.find({ status: "Completed" }).explain("executionStats")
{
  explainVersion: '1',
  queryPlanner: {
    namespace: 'lab5.visits',
    parsedQuery: { status: { '$eq': 'Completed' } },
    indexFilterSet: false,
    queryHash: 'F7CB0027',
    planCacheShapeHash: 'F7CB0027',
    planCacheKey: 'F1FC435E',
    optimizationTimeMillis: 0,
    maxIndexedOrSolutionsReached: false,
    maxIndexedAndSolutionsReached: false,
    maxScansToExplodeReached: false,
    prunedSimilarIndexes: false,
    winningPlan: {
      isCached: false,
      stage: 'FETCH',
      nss: 'lab5.visits',
      inputStage: {
        stage: 'IXSCAN',
        nss: 'lab5.visits',
        keyPattern: { status: 1, date: -1 },
        indexName: 'status_1_date_-1',
        isMultiKey: false,
        multiKeyPaths: { status: [], date: [] },
        isUnique: false,
        isSparse: false,
        isPartial: false,
        indexVersion: 2,
        direction: 'forward',
        indexBounds: {
          status: [ '['Completed', 'Completed']' ],
          date: [ '[MaxKey, MinKey]' ]
        }
      }
    },
    rejectedPlans: []
  },
  executionStats: {
    executionSuccess: true,
    nReturned: 9,
    executionTimeMillis: 1,
    totalKeysExamined: 9,
    totalDocsExamined: 9,
    executionStages: {
      isCached: false,
      stage: 'FETCH',

```

```

totalDocsExamined: 9,
executionStages: {
  isCached: false,
  stage: 'FETCH',
  nReturned: 9,
  executionTimeMillisEstimate: 0,
  works: 10,
  advanced: 9,
  needTime: 0,
  needYield: 0,
  saveState: 0,
  restoreState: 0,
  isEOF: 1,
  nss: 'lab5.visits',
  docsExamined: 9,
  alreadyHasObj: 0,
  inputStage: {
    stage: 'IXSCAN',
    nReturned: 9,
    executionTimeMillisEstimate: 0,
    works: 10,
    advanced: 9,
    needTime: 0,
    needYield: 0,
    saveState: 0,
    restoreState: 0,
    isEOF: 1,
    nss: 'lab5.visits',
    keyPattern: { status: 1, date: -1 },
    indexName: 'status_1_date_-1',
    isMultiKey: false,
    multiKeyPaths: { status: [], date: [] },
    isUnique: false,
    isSparse: false,
    isPartial: false,
    indexVersion: 2,
    direction: 'forward',
    indexBounds: {
      status: [ '['Completed', 'Completed'] ],
      date: [ '[MaxKey, MinKey]' ]
    },
    keysExamined: 9,
    seeks: 1,
    dupsTested: 0,
    dupsDropped: 0,
    peakTrackedMemBytes: 0
  }
},
},
},

```

```

queryShapeHash: '695D1066D6337A4E683805A7B034A70C3D58700E951CCD006C9451FF9262D219',
command: { find: 'visits', filter: { status: 'Completed' }, '$db': 'lab5' },
serverInfo: {
  host: 'DESKTOP-IKVG7C',
  port: 27017,
  version: '8.3.2',
  gitVersion: '89f54eb32e217d2f7bfcad50e5919f3033bb9611'
},
serverParameters: {
  internalQueryFacetBufferSizeBytes: 104857600,
  internalDocumentSourceGroupMaxMemoryBytes: 104857600,
  internalQueryMaxBlockingSortMemoryUsageBytes: 104857600,
  internalDocumentSourceSetWindowFieldsMaxMemoryBytes: 104857600,
  internalQueryFacetMaxOutputDocSizeBytes: 104857600,
  internalLookupStageIntermediateDocumentMaxSizeBytes: 104857600,
  internalQueryProhibitBlockingMergeOnMongoS: 0,
  internalQueryMaxAddToSetBytes: 104857600,
  internalQueryFrameworkControl: 'trySbeRestricted',
  internalQueryPlannerIgnoreIndexWithCollationForRegex: 1
},
ok: 1
}

```

```
lab5> // Переглядаємо створені індекси
| db.visits.getIndexes()
|
| // Видаляємо створений індекс
| db.visits.dropIndex("status_1_date_-1")
{ nIndexesWas: 2, ok: 1 }
```

```
lab5> db.visits.getIndexes()
[ { v: 2, key: { _id: 1 }, name: '_id_' } ]
```